

Comprehensive Study Guide: Special Senses, Endocrine, and Reproductive Systems

This guide integrates information from the PowerPoints and the hormone PDF.

I. Special Senses

A. Introduction

- General senses: widely distributed (touch, pressure, pain, temperature, vibration, proprioception).
- Special senses: localized, specialized receptors (vision, hearing, equilibrium, taste, smell).

B. Vision

1. Structures: Eyelids, conjunctiva, lacrimal apparatus, extrinsic eye muscles, eyeball (fibrous tunic – sclera, cornea; vascular tunic – choroid, ciliary body, iris; retina – rods, cones, macula lutea, fovea centralis).
2. Function: Refraction of light, accommodation, phototransduction (rods for dim light/ grayscale, cones for bright light/color).
3. Visual Pathway: Retina → optic nerve → optic chiasm → optic tract → thalamus → visual cortex.
4. Clinical: Myopia, hyperopia, astigmatism, cataracts, glaucoma, macular degeneration.

C. Hearing

1. Structures: Outer ear (pinna, auditory canal, tympanic membrane), middle ear (ossicles – malleus, incus, stapes; Eustachian tube), inner ear (cochlea, vestibule, semicircular canals).
2. Function: Sound wave transmission, amplification, conversion to nerve impulses. Cochlea differentiates pitch.
3. Auditory Pathway: Cochlea → vestibulocochlear nerve → brainstem → thalamus → auditory cortex.
4. Clinical: Conductive hearing loss, sensorineural hearing loss.

D. Equilibrium

1. Structures: Vestibule (utricle, saccule – static equilibrium), semicircular canals (dynamic equilibrium).
2. Function: Detecting linear acceleration (utricle/saccule) and rotational movement (semicircular canals).
3. Clinical: Vertigo, motion sickness.

E. Taste (Gustation)

1. Structures: Taste buds (papillae – circumvallate, fungiform, foliate, filiform).
2. Function: Detecting five primary tastes (sweet, sour, salty, bitter, umami). Different receptor mechanisms.
3. Gustatory Pathway: Taste buds → cranial nerves (VII, IX, X) → brainstem → thalamus → gustatory cortex.

F. Smell (Olfaction)

1. Structures: Olfactory epithelium (olfactory receptors, supporting cells, basal cells), olfactory bulb, olfactory tract.
2. Function: Detecting odorants (chemicals).
3. Olfactory Pathway: Olfactory receptors → olfactory bulb → olfactory tract → olfactory cortex.
4. Clinical: Anosmia.

II. Endocrine System

A. Introduction

- Endocrine vs. Nervous System: Hormones vs. neurotransmitters, slower/longer-lasting vs. rapid/short-lived effects.
- Hormone Types: Lipid-soluble (steroids, thyroid hormones) vs. water-soluble (peptides, proteins, amines).
- Hormone Actions: Synergistic, permissive, antagonistic.

B. Hypothalamus and Pituitary Gland

1. Hypothalamus: Controls pituitary gland, produces releasing and inhibiting hormones.
2. Anterior Pituitary: Produces GH, TSH, ACTH, FSH, LH, PRL. Regulated by hypothalamic hormones.
3. Posterior Pituitary: Stores and releases oxytocin and ADH (produced in hypothalamus).

C. Other Endocrine Glands

1. Pineal Gland: Melatonin (circadian rhythm).
2. Thyroid Gland: T3, T4 (metabolism, growth), calcitonin (lowers blood calcium).
3. Parathyroid Glands: PTH (raises blood calcium).
4. Thymus: Thymosin (immune function).
5. Adrenal Glands:

- Cortex: Aldosterone (sodium/water balance), cortisol (stress response), androgens.
- Medulla: Epinephrine, norepinephrine (fight-or-flight).
- 6. Pancreas: Insulin (lowers blood glucose), glucagon (raises blood glucose).
- 7. Gonads (Ovaries and Testes): Sex hormones (estrogen, progesterone, testosterone).

D. Clinical Applications:

- Diabetes mellitus, thyroid disorders (hypothyroidism, hyperthyroidism), adrenal disorders (Cushing's syndrome, Addison's disease), growth disorders.

III. Reproductive System

A. Male Reproductive System

1. Structures: Testes (seminiferous tubules, Leydig cells), epididymis, vas deferens, seminal vesicles, prostate gland, bulbourethral glands, penis.
2. Spermatogenesis: Process of sperm production.
3. Hormonal Control: GnRH, FSH, LH, testosterone, inhibin.

B. Female Reproductive System

1. Structures: Ovaries, fallopian tubes, uterus (endometrium, myometrium, perimetrium), cervix, vagina, vulva.
2. Oogenesis: Process of egg production.
3. Menstrual Cycle: Menstrual phase, follicular phase, ovulation, luteal phase. Hormonal regulation (GnRH, FSH, LH, estrogen, progesterone).
4. Clinical: Endometriosis, PCOS.

C. Pregnancy and Birth Control

1. Pregnancy: Hormonal changes (hCG, estrogen, progesterone).
2. Birth Control: Hormonal methods, barrier methods, IUDs, sterilization.

This comprehensive guide provides a structured overview of the key concepts. Use it in conjunction with the flashcards, practice questions, and interactive exercises for a more effective study experience. Remember to focus on understanding the connections between different systems and the clinical applications of the material.